Jesse Ramirez

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Technical Skills

Design/Analysis: Siemens NX, SolidWorks, CATIA, Onshape, Fusion 360/CAM, Revit, HyperMesh, Patran, Nastran, ANSYS, MATLAB
Programming/Test: Python, C++, LabVIEW, DAQ, thermocouples, load cells, instrumentation, P&IDs, fluid-system integration
Manufacturing: CNC/manual milling, manual lathe, drill press, FDM, metal DMP/DFM, waterjet, laser cutting, welding, soldering

Education

University of California, Berkeley — May 2026
B.S. Mechanical Engineering; Minor in Aerospace — Berkeley, CA

- **Relevant Coursework:** Thermodynamics, Heat Transfer, Fluid Mechanics, Combustion, Composite Materials, Flight Mechanics, Dynamics & Controls, Solid Mechanics, Mechatronics Design, Manufacturing

Engineering Experience

The Boeing Company — May 2026 – Present
Structural Analysis Engineer — *Everett, WA*

- Created and modified finite-element models of 777X interior structures using HyperMesh and Nastran, applying combined seat-loading cases and reviewing stress results against supplier specifications.
- Independently evaluated supplier structural analyses using hand calculations and finite-element results, checking static strength, buckling, and yield and ultimate margins for consistency with engineering requirements.

Southland Industries — May 2025 – Aug. 2025
Design Engineering Intern — *Union City, CA*

- Built an Excel cooling-load matrix for six Google data-center laboratories, enabling comparison of server-rack power demand and air-, water-, and hybrid-cooling requirements by equipment type.
- Produced load calculations, Revit layouts, piping designs, and construction drawings for a 4,250-ft^2 tenant improvement, revising mechanical systems as server quantities and heat loads changed.

Space Technology and Rocketry (STAR) — Sep. 2023 – May 2026
Propulsion Lead — *Berkeley, CA*

- Designed, manufactured, and tested ablatively and regeneratively cooled LOX/RP-1 engines; resolved instrumentation, leakage, and ignition failures before a successful launch and recovery to approximately 8,000 ft.
- Designed a 4.79-kN, 540-psi Inconel 718 regenerative thrust chamber, sizing cooling channels against thermal, boiling, pressure-loss, and 0.45-mm DMP manufacturing constraints with 3D Systems.
- Led propulsion requirements, design reviews, procurement, testing, and launch operations while owning the P&IDs and integration of pressure-regulated vehicle and ground fluid systems.

Berkeley Fire Research Lab — Feb. 2025 – May 2026
Undergraduate Researcher — *Berkeley, CA*

- Supported NASA's SoFIE-MIST investigation aboard the International Space Station by conducting 10–15 PMMA combustion tests per session to study flame spread and limiting oxygen concentration under spacecraft-relevant conditions.
- Varied oxygen concentration, chamber pressure, preheat temperature, applied heat flux, ignition conditions, and specimen diameter to generate experimental data for evaluating microgravity flammability models.
- Built an actuated sample fixture and instrumented the burn rig with thermocouples, oxygen sensors, LabVIEW DAQ, and video imaging; lathe-machined 3.20-mm PMMA specimens to approximate thermally thin behavior.

Selected Projects

Spider Seeder Reforestation Robot | *Onshape, C++, ESP32* — Spring 2026

- Designed and assembled the drill, servo-driven linkages, electronics housing, and sensor mounts in Onshape for a six-legged reforestation robot.
- Programmed an ESP32 state machine for 18 servos, drilling, seed dispensing, and safety sensors; resolved motor stalls, overheating, and power-distribution failures. Earned the Electromechanical Design Award.

Bio-Inspired FDM Composite Thermal Tiles | *ANSYS, MATLAB, DAQ* — Spring 2025

- Designed, FDM-manufactured, and tested eight PA6 and carbon-fiber PA6 configurations, comparing four geometries using transient ANSYS models, Type-K thermocouples, infrared imaging, and DAQ.
- Identified carbon-fiber PA6 scales as the most deformation-resistant configuration, sustaining 76 seconds of heat-gun exposure with minor deformation versus 20–27 seconds for the spike configurations.